Vidit Arora

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SDE and AI/ML intern candidate at DTU (CGPA 9.6/10). I build multi-agent large language model (LLM) systems, retrieval-augmented generation (RAG) pipelines, harnesses, AI skills, and full-stack apps.

Education

Delhi Technological University (DTU), New Delhi *Aug 2025 - May 2029*
B.Tech in **Software Engineering** | CGPA: **9.6/10** | Department **Rank 3**
Class XII **(CBSE)**, Daffodils Public School: **94.0% (2025)** | Class X **(ICSE)**, Hartmann College: **94.6% (2023)**

Technical Skills

LLM agents and RAG: LangGraph, multi-agent systems, RAG, ChromaDB, pgvector, Google AI Studio, prompt, context & harness engineering, citation verification, ablation study, System One models like Jev and Laya. *(Research Console, Pulse, Pluvial-AI, Opening Radar, Feynman Twin)*
Backend and deployment: FastAPI, PostgreSQL, SQLAlchemy, Redis, SQLite, Next.js, React, Tailwind, Vercel, Streamlit, Gradio, Git, Linux (*Pulse, Opening Radar, Vaani-Setu)*
Deep learning and speech: PyTorch, Mamba state-space models, Hugging Face Transformers, sentence-transformers, ONNX Runtime (INT8 models) (*Mamba-Dexformer, Vaani-Setu, Deep Learning Specialization)*
Classical ML and forecasting: scikit-learn, pandas, NumPy, time-series forecasting (*Kaggle campus competitions, Machine Learning Specialization)*
Languages: Python, C, C++, Java, JavaScript, TypeScript, SQL

Experience

Research Intern, Agentic AI Security *Jun 2026 - Present*
AIMS Lab (research lab of the Department of Information Technology), DTU | Advisor: Prof. D.K. Vishwakarma

- Studied how multi-step LLM agents fail under adversarial input, including weaknesses in their chain-of-thought reasoning, to support agentic security evaluation.
- Reviewed agentic AI threat-model literature published from 2022 to 2026 and built an exploratory study of prompt-injection recognition in agent reasoning (GitHub).

Projects

Agentic Deep Research Console GitHub
Python, Gemini API, ChromaDB, Streamlit

- Built a multi-agent RAG system that answers research questions over **700+ AI/ML and CS arXiv papers** (2024 to 2026). A Planner splits each question into sub-searches; a Reflector critiques the evidence and triggers new searches, and a Synthesizer writes the answer.
- Wrote a Citation Verifier that enforces exact [arxiv_id] inline citations and drops claims that lack one, to reduce unsupported claims.
- Made the pipeline idempotent and resumable through API rate limits. Added a **30-question ablation harness** that switches each agent on or off to measure what it adds.

Injection Recognition in Agent Reasoning GitHub
Python, AgentDojo, Ollama

- Built a pipeline that reads an agent's own reasoning to tell if it noticed a prompt injection or ignored it. Ran 3 local open-weight models (DeepSeek-R1 14B, Qwen 2.5 14B and 7B) on 2 AgentDojo suites, banking and workspace.
- Checked the recognition classifier against a human annotator on a 60-trace sample: Cohen's kappa 0.75 (3-class, in-sample).
- Observed that resistance was mostly blind: in the banking main pools no model showed recognition in its reasoning (exploratory, 31 to 42 runs per model).

Pulse: Agentic Social Media Manager GitHub
LangGraph, FastAPI, PostgreSQL (pgvector), Redis, Next.js

- Built **five LangGraph agents** (Scout, Strategist, Copywriter, Guardrail, Sentinel) on FastAPI, async SQLAlchemy and PostgreSQL. They find trends, time posts, draft copy, check brand voice, and triage comments.
- Added a brand-voice guardrail that scores each draft by cosine similarity to the brand's embedding centroid, with up to 2 retries. Added a Sentinel circuit breaker that pauses scheduled posts when the 30-minute share of negative comments crosses a threshold.
- Retrieved top-quartile past posts with pgvector similarity search for few-shot drafting. Recommended posting times from an engagement heatmap with a LinUCB contextual bandit scaffold.

Pluvial-AI: 1st Prize, AIMS x Mireye Internal Hackathon GitHub

Team of 3 | Python, FastAPI, PostgreSQL, MapLibre

- Proposed the idea and wrote the agent layer: Investigator, Skeptic and Adjudicator agents that judge ground risk for a US address from live Mireye data and cite a sampled map point for every claim.
- Team backtest in triage mode on 50 cases drawn from **395,783 Houston 311 complaints**: **78.6% precision at 30.6% recall**. The system may return 'unresolved' when evidence is weak.

Opening Radar Live | GitHub

Next.js, React, TypeScript, Tailwind, Vercel, TinyFish

- Built a live job-search agent app that searches **5 hiring platforms** (Greenhouse, Lever, Ashby, Workable, SmartRecruiters) and reads up to 20 postings in parallel with TinyFish.
- Wrote a rule-based ghost-job check and a decision model with level and location gates that rank postings.
- Added a daily monitor for new postings and application-form prep that never types or submits anything for the user.

Mamba-Dexformer: Multi-Speaker Speech Separation GitHub

PyTorch, Mamba, Gradio

- Built a speech separator (1D convolutional encoder, 4 Mamba state-space blocks, 2 ConvTranspose decoders) that extracts **up to 5 overlapping speakers**, one per pass. Trained it with OR-PIT loss for 115 epochs on a curriculum that grows from 2 to 5 speakers.
- Found the 2-speaker score falling in later curriculum phases, consistent with catastrophic forgetting, and proposed mixed-batch training as a control.
- Published the Kaggle dataset speech-sep-mixtures (LibriMix extended to 4 and 5 speakers). DataSet

Vaani-Setu: Offline Literacy App (SIH 26) GitHub

SIH Team | ONNX Runtime, FastAPI, Next.js

- Co-built an offline literacy and numeracy app for Jharkhand's tribal (Santali, Ho and Mundari) students (PALASH programme, Problem Statement 42) designed for **2 GB RAM** Android tablets.
- The app runs Hindi speech recognition, translation (IndicTrans2) and speech synthesis on the device as INT8 ONNX models.

Richard Feynman Digital Twin Live | GitHub

LangGraph, ChromaDB, Gemini, ElevenLabs, Streamlit

- Built a live RAG agent over Feynman's lectures, books and letters that answers by text or voice, writes and runs Python to draw Feynman diagrams, and keeps long-term memory in SQLite.

Achievements

- **Top 10** finalist (team), ISRO Bharatiya Antariksh Hackathon 2026 (PS-13), on satellite imagery.
- **Winner** (team), BITS at Invictus '26, DTU: three stages covering competitive programming, CS fundamentals and an algorithmic treasure hunt.
- **Rank 2** of 47 in an intra-department Kaggle time-series forecasting competition, and Rank 2 of 187 in a campus-wide Kaggle ML competition.
- **1st prize** (team of 3), AIMS x Mireye (YC s26) internal hackathon, with Pluvial-AI.

Leadership and Activities

Co-Head, AI & ML Society (AIMS), DTU *Aug 2026 - Present*

- Teach machine learning to juniors with fellow co-heads: each session has 100+ students, and 20+ sessions are planned for our juniors.
- Hosted the AIMS Orientation for the 2030 batch on the auditorium stage, in front of 500+ attendees.
- Volunteered in crowd, event and backstage management at the AWS Build with Bharat offline day and the AIMS x TinyFish workshop and Buildathon. Earlier an AIMS ML team member (Feb 2026 - Sep 2026).

Team Lead, Fresources @DTU *Jul 2026 - Present*

- Lead the management of the software engineering department's study resources and keep a 4,000+ student community up to date with campus events and notices. Member from Feb 2026 - Jun 2026.

Society of Software Engineers (SSE), DTU *Oct 2025 - Jul 2026*

- AI/ML team member, data structures and algorithms (DSA) and competitive programming (CP) team member.
- Research Assistant on music source separation (Mar 2026 - Jul 2026). PR and design team member where I promoted and volunteered the society's flagship events at Invictus '26.

Certifications

- Machine Learning Specialization (Stanford Online and DeepLearning.AI, Coursera), Dec 2025 Certificate
- Deep Learning Specialization (DeepLearning.AI), Mar 2026 Certificate
- Retrieval Augmented Generation (DeepLearning.AI), Jul 2026 Certificate